

GUSTAVO LUJAN CENTENO

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EDUCATION

University of Massachusetts Lowell Lowell, MA
Bachelor of Science in Computer Engineering May 2026
GPA: 3.33/4.00

University of Massachusetts Lowell Lowell, MA
Master of Science in Computer Engineering (in progress, part-time)
Graduate GPA: 4.00/4.00

EXPERIENCE

Graduate Research Assistant | UMass Lowell, Lowell, MA June 2026 – Present

- Conducting research under Prof. Seung Woo Son on porting a netlist fault injection framework to AWS EC2 FPGA (F2) instances.
- Analyzing existing Xilinx VMK180-based framework to identify porting requirements for the AWS F2 Custom Logic (CL) environment.
- Gaining hands-on experience with cloud FPGA computing platforms, Vivado/Vitis toolchains, and the AWS FPGA Developer AML.

SKILLS

Programming: C/C++, C#, Python, MATLAB, Go, ARM Assembly, VHDL

Hardware: AVR/Arduino, Raspberry Pi Pico, FPGA (DE10-Lite, DE1-SoC), Analog Circuits, Op-Amps, Soldering

Tools: Vivado, Quartus, Vitis, Analog Discovery, Multisim, LTSpice, Git, Arduino IDE, SolidWorks

Platforms: AWS EC2 F2, Xilinx UltraScale+, Xilinx VMK180, Linux

Languages: English, Spanish, Italian

RELEVANT COURSEWORK

Operating Systems · Digital Systems & FPGA Design · Microprocessors & Embedded Systems · Computer Architecture · Network Design · Electronics & Circuit Design · Software Engineering · Hardware Security

PROJECTS

Fabrication Machine for Microelectrode Arrays — UML Capstone Spring 2026

- Contributed to automated MEA fabrication system as part of a multidisciplinary engineering team.
- Worked on hardware-software integration, motion control, and testing of embedded subsystems.
- Performed iterative testing, calibration, and debugging to improve system performance and automation.

LoRa Mesh Network Optimization — UML Research Spring 2026

- Extended a Python simulation by implementing Grey Wolf Optimizer (GWO) and PSO routing algorithms on LoRa mesh networks.
- Designed test sequences measuring latency, energy consumption, and delivery rate across 5-100 node configurations.
- Validated results on real RAK LoRa hardware running Meshtastic firmware.

FPGA Temperature Sensor — Clock Domain Crossing — UML Project Fall 2025

- Designed a VHDL system on DE10-Lite reading the on-chip temperature sensor and displaying live readings on seven segment displays.
- Implemented async FIFO with Gray code synchronizers to safely transfer data between 1 MHz ADC and 50 MHz display domains.
- Built a custom VHDL clock divider replacing the deprecated PLL megafunction in Quartus Prime Lite 24.1.

Cooperative Real-Time Scheduler — UML Project Fall 2025

- Implemented a cooperative real-time task scheduler in C/C++ on an AVR microcontroller using the Arduino Uno board.
- Compared Round-Robin vs Priority scheduling measuring latency, deadline misses, and CPU usage across 3 task types.
- Designed scheduling logic, timing counters, and task execution control for multiple concurrent embedded tasks.

CERTIFICATIONS

IBM – Machine Learning for Data Science Certificate